

Spoken Tutorial – Understanding Retrieval-Augmented Generation (RAG)

Title: Understanding Retrieval-Augmented Generation (RAG)

Module	Advanced AI Concepts
Episode	Introduction to RAG
Learning Objective	At the end of this tutorial learner will be able to 1. By the end of this tutorial, you will be able to explain what Retrieval-Augmented Generation (RAG) is and how it works.
Approx. Duration	10-12 minutes
Outline	• Introduction to the 'closed-book' problem in AI • What is Retrieval-Augmented Generation (RAG)? • Understanding the 'Retrieval' component • Understanding the 'Generation' component • The 3-step RAG flow: Query, Retrieve, Generate • Benefits of using RAG for AI applications
Meta Tags	RAG, Retrieval-Augmented Generation, LLM, Large Language Models, AI, Generative AI, Knowledge Base, Chat with your data, Spoken Tutorial
Pre-requisite Tutorial	Basic understanding of Large Language Models (LLMs) and their working.

Script

Visual Cue	Narration
	Hello and welcome to this Spoken Tutorial on Retrieval-Augmented Generation , or RAG . Here, we will learn how RAG helps AI models give better answers.
	By the end of this tutorial, you will learn what Retrieval-Augmented Generation , or RAG , is. You will also understand how RAG works.
	To follow this tutorial, you need a computer and good internet. You do not need to install any special software for this lesson.
	Before you start, you should know what Large Language Models , or LLMs , are. You should also know how they write text. This will help you understand RAG better.

An AI chatbot looking confused, with a thought bubble showing vague or incorrect information.	Have you seen an AI give wrong answers, especially for new information? This happens because the AI answers only from what it already knows. It is like a student taking a 'closed-book' exam.
A student happily taking an exam with an open book next to them, contrasting with another student struggling with a closed book.	Retrieval-Augmented Generation , or RAG , solves this 'closed-book' problem. It is a simple idea that helps AI models give much better answers. Think of it like an 'open-book' exam.
A large language model brain icon with arrows pointing to external knowledge sources like books and documents, showing integration.	RAG connects outside knowledge with the AI model. It gives the AI new facts to read before it answers your question. This helps the AI give correct and full answers.
A friendly librarian character searching through a digital library database, with various documents and PDFs floating around.	The 'Retrieval' part of RAG is like a smart Librarian. When you ask a question, it searches a special knowledge base. This knowledge base can be your PDF files, company documents, or other papers.
A digital hand pointing to highlighted relevant paragraphs within a stack of documents, indicating precise retrieval.	The Librarian finds the exact parts of documents that match your question. It quickly gets the best information for the AI . This is the first important step in RAG .
An AI chatbot character (Eloquent Expert) receiving two inputs: a speech bubble with a question and a stack of documents.	The 'Generation' part is the AI model itself, like an Expert. It gets two things: your question and the useful documents from the Librarian.
The AI chatbot character confidently answering a question while pointing to the provided documents as its source, not its head.	Now, the Expert does not answer from its own memory. It is told to answer your question by using only the documents given to it. This makes sure the answer is based on facts.
A clear flowchart showing three connected boxes: 'Query' -> 'Retrieve' -> 'Generate', with an arrow looping back from 'Generate' to 'Query'.	RAG works in 3 simple steps. Step 1: You ask a question. Step 2: The system finds information. Step 3: The system gives you an answer. Many 'Chat with your data' apps use this method.
An AI chatbot providing an answer with a small citation icon or footnote, next to a checkmark indicating accuracy.	RAG makes the AI give true and clear answers. The AI can now show its sources. This helps you trust the AI 's answers more. It gives better and more useful information.
	In this tutorial, we learned about Retrieval-Augmented Generation , or RAG . RAG helps AI models give better answers. It does this by finding outside information and then creating a response. This makes answers more correct and easy to check.

	For your assignment, find an example of a 'Chat with your data' app that uses RAG. Try to see how the finding and creating parts work in that app.
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